SECTION 01 11 00

SUMMARY OF WORK & BUREAU PROCEDURES

PART 1 GENERAL

1.01 SUMMARY OF WORK

A. All work described in these documents and all work necessary to complete the work as described or shown is to be executed in a thoroughly substantial and workmanlike manner. All work shall be done by persons who are thoroughly experienced in their particular trade or craft. The work includes, but is not necessarily limited to the following description:

FCC Terre Haute proposes for approximately 17,000 square feet of asphalt roof to be removed and a new metal roofing system installed. Prior to roof system installation repairs will be made to the rafters and decking in the areas affected by the roof leaks in FPC HVAC, Plumb. and Comm Shop in preparation for roof repairs.

Project Name: *Project 24Z4AT8 – Replace Roof FPC HVAC, Plumb. And Comm Shop*
Location: *FCC Terre Haute*
Address: *4200 Bureau Road Terre Haute, Indiana 47802*
Owner: *U.S. Department of Justice, Federal Bureau of Prisons (FBOP)*

1.02 CONTRACTOR USE OF PREMISES

A. The Contractor shall limit its use of the premise to the work indicated, so as to allow for Government occupancy and use. During the entire construction period, the Contractor shall coordinate use of the premises for construction operations with the FBOP Contracting Officer's Representative (COR).
   1. All visitors to the construction site shall register with the FBOP site officials. The Contractor shall not allow employees to have personal visitors on site.
   2. The project site will be open to the Contractor from Monday through Friday, with exception of federal holidays, for the duration of the project. The project schedule is to be based on a normal 40 hour work week. The Contractor shall establish a normal work week schedule for the duration of the project.
      a. Work outside the secure perimeter of the institution will be allowed between the hours of 6:30 AM to 4:30 PM. This will allow the Contractor's work force to arrive on-site in the morning and prepare for entering the secure perimeter of the institution. It will also allow the Contractor's work force to button-up their lay-down/staging area after exiting the secure perimeter at the end of the work day.
      b. Work inside the secure perimeter of the institution will be allowed between the hours of 7:30 AM to 3:00 PM. Work inside the secure perimeter is restricted to the operation of the institution's rear gate. The Contractor's work force will be allowed to enter the rear gate at approximately 7:30 AM each workday. At this time, the FBOP staff will begin personnel and tool security check-in procedures. Once the check-in procedures are completed, the Contractor will be escorted by an FBOP escort to the work area. The rear gate closes at 3:00 PM; therefore, the Contractor's work force will be escorted back to the rear gate by FBOP staff no later than approximately 2:30 PM, in order to allow enough time for the completion of security check-out procedures prior to exiting the secure perimeter.

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3. Contraband: The Contractor shall inform his/her employees that intoxicating beverages, drugs, weapons, etc.., will not be allowed on the site. Introduction of contraband shall be subject to the penalties prescribed by Title 18 USC 1791.
4. Explosives: The use of explosives is absolutely forbidden without written authorization from the FBOP Contracting Officer upon no less than thirty (30) days advance written request.
5. Cell phones, pagers, and radios are not allowed within the secure perimeter.
6. Parking: Parking areas, either on or off-site will be furnished for workmen by the Contractor at a location approved by the FBOP Contracting Officer's Representative (COR).
7. Temporary Office: Any temporary field office used by the Contractor shall be located in an area approved by the FBOP Contracting Officer's Representative (COR).
8. The Contractor shall review his or her site layout plan for staging, stockpiles, etc.., with the FBOP Contracting Officer's representative (COR). The Contractor shall maintain all staging and construction areas in a neat and orderly fashion.
9. The Contractor shall provide protection of the work against weather, rain, wind, storms, frost, cold or heat so as to maintain all work, materials, apparatus and fixtures free from injury or damage.

1.03 CONTRACTOR PERSONNEL AND EQUIPMENT SECURITY

A. Contractor employees working at a FBOP facility will be subject to a background investigation after award of the contract. Contractor employees must complete the applicable forms provided by the FBOP Contracting Officer's Representative. Following the investigation, the FBOP will provide the Contractor with a list of employees approved to work at the institution. If a Contractor's employee is not approved, only that fact will be communicated to the Contractor. No additional justification will be released, in compliance with Privacy Act requirements.

B. Contractor employees working within the secure perimeter shall wear a clearance badge issued by the institution at all times, and shall be escorted by an institution staff member.

C. Tools brought into the institution's secure perimeter will be inventoried at the beginning and end of each work day by FBOP staff.
   1. All tools used by the Contractor working within the secure perimeter shall be under constant supervision. Loss of a tool shall be reported immediately to the escorting FBOP staff.
   2. No explosive action tools shall be used on the property at any time.
   3. Materials brought inside the secure perimeter of the institution shall be limited to a one-day supply. Surplus materials that are not installed in one work day must be moved outside the secure perimeter at the end of the day.

D. Contractor employees working on the FBOP site shall be required to undergo a general institution orientation on responsibilities, procedures and laws governing conduct when working at an FBOP institution. These rules include, but at not limited to the following:
   1. Weapons and ammunition of any kind are not allowed on Federal property, including the parking lot.
   2. Do not give, or offer to give, anything to an inmate.
   3. Do not bring any correspondence into or out of the institution.
   4. Do not accept gifts or favors from inmates.
   5. Leave personal items locked in your vehicle or secure them at the Contractor's lay-down area.
   6. A form of photo ID is required to enter the facility.
   7. Revealing clothing is not acceptable attire and work attire is required at all times.

8. Conduct yourself in a responsible, mature manner.
9. Report an unusual action(s) or behavior to FBOP staff.

1.04 EXECUTION OF WORK AND CLARIFICATION OF INFORMATION

A. It is the intention of the Construction Documents to provide a complete project. The Contractor shall be responsible for this result. The Contractor shall verify all measurements and be responsible for the correctness of same. The Contractor shall layout their work and be responsible for all lines, elevations, and measurements of buildings, utilities, and other work executed by them. This includes verification of existing conditions, in any, that affect their work or to which their work is to be fitted.

B. The Contractor shall carefully study and compare Contract Documents before proceeding with fabrication and installation of work. Coordinate scheduling, submittals and work to ensure efficient and orderly sequence of installation.

C. Should it appear that the work intended to be described is not sufficiently detailed or explained by the Construction Documents, the Contractor shall issue a Request for Information (RFI) to the FBOP Contracting Officer. RFI's shall be used by the Contractor to clarify information in the Construction Documents only. Each RFI shall be numbered sequentially, dated and indicate a reasonable date for the Contracting Officer's written reply. All RFI questions shall be thoroughly researched by the Contractor, prior to presenting them as RFI's. In no case shall any work proceed in uncertainty.

1.05 DEFINITIONS

A. Contracting Officer: The officer who executes the contract on behalf of the United States; this shall include a duly appointed successor. The Contracting Officer is the only official authorized to administer, modify, and/or terminate the contract. The Contracting Officer makes related approvals, determinations, and findings pertaining to the contract and is the only individual on the project staff who can obligate funds on behalf of the Government.

B. FBOP Contracting Officer's Representative (COR): The FBOP representative located at the construction project site or authorized representative(s), who has overall responsibility (except for contractual matters) for the administration of the operations and activities at the construction project site. References to "Government Representative" or "Project Representative" shall mean FBOP Contracting Officer's Representative (COR). The COR represents the Contracting Officer for all technical matters that arise under the contract. The COR shall be specifically responsible for the following areas as they relate to the administration of the construction contract. This delegation of duties and responsibilities does not preclude the Contracting Officer form also performing these duties.
   1. Monitor and review the progress of the work. This will include the conducting of onsite observations of the work as it is in progress. The COR shall issue reports concerning deficiencies and/or omissions observed in the work.
   2. Receive submittals to verify conformation with the Contract Documents.
   3. Review requests for information from the Contractor. This does not include the authority for any change that may result from the approval of the request for information. Authorization for any change can come only from the Contracting Officer.
   4. Monitor that all tests required by the Contract Documents are performed. This may include the documentation of the test procedures and results.
   5. Review sample construction and job site mockups for conformance with the Contract Documents.

6. Review and sign the application and certificates for payment as submitted by the Contractor. Approval of the certification must be made by the Contracting Officer.
7. Evaluate the Contractor's performance and prepare a performance report at the completion of the Contract.
8. Review and recommend final approval of the Contractor's schedule for the Contracting Officer as required by the Contract Documents. This process would include the Schedule of Values.
9. The COR will approve the placement of the Contractor's trailers and general administration of the construction site as it relates to the construction work.
10. The COR shall not, and has not authority to:
    a. Re-delegate any authority to any other person.
    b. Change any of the terms of the Contract or sign any modification to the Contract.
    c. Obligate the payment of any money by the Government.
    d. Cause the Contractor to incur cost not otherwise covered by the Contract.

C. Contractor: The sole proprietorship, partnership, joint venture, corporation, or other legal entity that agrees to provide all labor, material, and services required in the contract. In order to administer and supervise the Contract Work, the Contractor shall provide the following minimum supervisors:
1. Contractor's Project Manager: The Contractor's representative experienced in administration, supervision and quality control of all construction work required by the Contract. This includes, but is not limited to enforcing all applicable federal, state and local standards.
2. Project Field Superintendent: The Contractor's representative experienced in general field supervision of all construction work required by the Contract. Supervise, direct, inspect and coordinate the construction work of the Contractor, subcontractors, suppliers and installers, and to assure compliance with the Construction Schedule.

D. Subcontractor: The term, as employed herein, includes a business having a direct contract with the Contractor, and it includes one who furnishes material according to the Drawings and Specifications of this work. This definition also includes businesses having contracts with subcontractors. Subcontractors are considered to be Contractor employees.

E. Approve: Only the Contracting Officer can approve or disapprove Contract actions. Even if the Specifications indicate that an individual other than the Contracting Officer (such as the "FBOP Contracting Officer's Representative") will approve or disapprove an action, it is understood that only the Contracting Officer has this authority. In no case will "approval" by the Contracting Officer be interpreted as a release of the Contractor from responsibility to fulfill the requirements of Contract Documents

F. Industry Standards and General Applicability of Standards: Except where more explicit or more stringent requirements are written directly into the Contract Documents, applicable standards of the construction industry have the same force and effect (and are made a part of Contract Documents by reference) as if copied directly into Contract Documents, or as if published copies were bound herein. Any materials, equipment, or workmanship specified by references to published manufacturer's specifications, industry standards, building codes, or other referenced published standard shall comply with the requirements of the latest editions or revisions thereof and any amendment or supplements in effect on the date of the Invitation for Bid, except as limited to type, class, grade, or modification in the Specification. The Contractor, if requested, shall furnish an affidavit from the manufacturer certifying that material or product delivered to the project site meets requirements specified.

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1. "Federal Specification", "Commercial Standards" and "Simplified Practice Recommendations: can be purchased from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.
2. Standards of associations, institutes or similar organizations referred to in the Specifications may be obtained directly from the organization.

G. Substantial Completion: The stage in the progress of the work or designated portion thereof is sufficiently complete in accordance with the contract documents so the Government can occupy or utilize the work for its intended use.

1.06 PROJECT MEETINGS

A. Pre-Construction Meeting: The Contracting Officer shall schedule and conduct a pre-construction conference and organizational meeting at the project site or other convenient location prior to or at the time of receipt of the Notice to Proceed for the Construction Contract. Contracting Officer shall arrange for a record of the meeting to be taken and distributed to all concerned. Instructions given that are supported in the Contract Documents are considered binding on all contracting parties.
   1. The attendees shall include: the Contracting Officer, the FBOP Contracting Officer's Representative, the Contractor's Project Manager, the Project Field Superintendent, and major subcontractors and other technical support disciplines as appropriate. Attendees will also include representatives from the correctional institution, as deemed necessary by the Contracting Officer.
   2. The meeting itinerary shall include, but not be limited to issues such as:
      a. Introductions and coordination of the various parties in attendance.
      b. Use of premises and security procedures.
      c. Construction scheduling and work sequencing.
      d. Processing of shop drawings, product data, samples, request for information, contract modifications, and applications for payment.
      e. Contract closeout requirements.

B. Construction Progress Meetings: The Contracting Officer and the COR shall co-conduct progress meetings at the project site on a monthly basis, or at intervals as determined by the Contracting Officer.
   1. The attendees shall include:
      a. FBOP: Contracting Officer, COR, Institution Representatives, and technical support disciplines as appropriate.
      b. Contractor: Project Manager, Project Field Superintendent, and technical support disciplines as appropriate (Subcontractors, Manufacturers, Suppliers, etc…).
   2. The meeting itinerary shall include, but not be limited to issues such as:
      a. Progress since previous meeting, and completion date confirmation.
      b. Field observations and the construction schedule.
      c. Corrective measures necessary to maintain the project standards, resolve current problems, or head-off anticipated conflicts.
      d. The status of: project submittals, request for information, and contract modifications.

1.07 CONSTRUCTION SUBMITTALS

A. The types of submittal requirements specified in these requirements include shop drawings, product data, samples, certificates of conformance or compliance, certified test or inspection reports, and miscellaneous work-related submittals for construction. Individual submittal requirements are specified in applicable Sections for each unit of work. Provide a standard transmittal form with each submittal.

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B. The Contractor shall ensure submittals (product data, shop drawings, etc.) produced by subcontractors, vendors, or other entities are reviewed for adequacy of coordination and meeting the design intent of the construction documents prior to the submission to the Government. The Contractor shall stamp and certify that each submittal was reviewed, approved, and coordinated with the requirements of Work and the Contract documents. Failure on the part of the Contractor to indicate review and approval on submittals, prior to submission to the COR, will result in the submittals not being accepted and returned to the Contractor. The COR shall return the submittals to the General Contractor within approximately fifteen (15) calendar days.

C. Preparation of Submittals: Provide permanent marking on each submittal to identify project, date, Contractor, Subcontractor, submittal name, Specification section number, and similar information to distinguish it from other submittals. Number resubmittals with original number and an alphabetic suffix.

D. Coordination and Sequencing: Coordinate preparation and processing of submittals with performance of the work so that work will not be delayed by submittals. Coordinate and sequence different categories of submittals for the same Work, and for interfacing units of Work, so that one will not be delayed for coordination of review with another. The Contractor shall promptly notify the Contracting Officer and the COR when submittal action is delayed for any reason, citing the cause for delay.

1.08 CONSTRUCTION CONTRACT MANAGEMENT REQUIREMENTS

A. Construction Scheduling: The Contractor shall provide a detailed Construction Schedule with activity entries for each portion of the construction work, in chronological order, cumulatively indicating the entire Contract Performance Period. The Construction Schedule shall include the initial detailed construction schedule, and all subsequent construction schedule updates and revisions, as required over the life of the contract.

1. The entire contract performance period as established by the Contract shall establish a Final Contract Completion Date and shall be utilized in the planning and presentation of the Contractor's Construction Schedule. As the Construction Schedule is being developed, the Contractor may project an earlier completion date. The government may elect to accept a schedule that projects an earlier completion date; however, this acceptance shall not modify the Final Contract Completion Date. The time difference between the Contractor's projected completion date and the Final Contract Completion Date shall be owned and for the mutual use of both the Contractor and the Government. The Government reserves the right not to accept any schedule it determines to be unrealistic in length.
2. The purpose of the Construction Schedule shall be to:
   a. Assure adequate planning, staffing, scheduling and reporting during execution of the work by the Contractor.
   b. Assure coordination of the work among the Contractor, subcontractors, manufacturers and suppliers.
   c. Assist the Contractor and Contracting Officer in monitoring the progress of the work and evaluating proposed changes to the Contract and the Construction Schedule.
   d. Assist the Contracting Officer and the Contractor in the preparation and evaluation of the Contractor's monthly progress payments.
3. The Construction Schedule shall be updated on a monthly basis throughout the entire Contract time and until Project Completion. The Contractor and the Contracting Officer shall review the actual progress made during each Construction Progress Meeting. The Contractor shall inform the Contracting Officer of any current or anticipated conditions

which may delay the work, and provide any additional written information necessary to support these conditions. The Contractor shall update the Construction Schedule to reflect the actual progress of the project, as of the date directed by the Contracting Officer. If it appears the Construction Schedule no longer represents the actual prosecution and progress of the work, the Contracting Officer will direct the Contractor to revise the Construction Schedule.

4. The Construction Schedule shall:
   a. Show the sequence of construction by activity, with dates for beginning and completion of each element of construction.
   b. Identify each item by specification section number.
   c. Show accumulated percentage of completion for each item.
   d. Coordinate with the Schedule of Values (See Item B, below).
   e. Provide sub-schedules to define critical portions of the entire Construction Schedule
   f. Include a narrative report to discuss issues of importance, such as, but not limited to: anticipated delays, contract modifications, revisions, etc.

B. Schedule of Values: The Contractor shall prepare a Schedule of Value (SOV) for the total
   1. The Schedule of Values shall include:
      a. Project name, location and contract number.
      b. Contractor's name and address.
      c. Date of submittal.
   2. The SOV shall be arranged in a tabular form with separate columns to indicate the following for each item listed:
      a. Generic name.
      b. Related specification section, if applicable.
      c. Name of subcontractor, if applicable.
      d. Name of manufacturer or fabricator, if applicable.
      e. Name of supplier, if applicable.
      f. Change orders (modifications) that have affected value, if applicable.
      g. Dollar value.
      h. Percentage of Contract sum rounded to the nearest percent, adjusted to total 100 percent.
   3. The SOV breakdown shall breakdown the Contact sum in sufficient detail to facilitate continued evaluation of applications of payment and progress.
   4. SOV amounts shall be rounded to the nearest whole dollar. The total of all SOV amounts shall equal the Contract sum.

C. Contractor Applications for Payments: The Contracting Officer will provide the Contractor with his/her requirements for submitting applications for payment. The application for payment may include, but is not limited to: lien waivers for work paid to date, receipts or vouchers for payment of labor, budget cost report, listing stored materials, and any other supporting data requested by the Contracting Officer.
   1. Initial Application for Payment: Actions and submittals that must precede or coincide with submittal of the first application for payment include the following:
      a. List of subcontractors.
      b. List of principal suppliers and fabricators.
      c. Schedule of Values.
      d. Contractor's construction schedule.
      e. Submittal schedule.

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2. Final Payment Application: Actions and submittals that must precede or coincide with submittal of the final application for payment include the following:
   a. Completion of project closeout requirements.
   b. Completion of items specified for completion after substantial completion.
   c. Transmittal of Record Documents and other required project construction records to Government.
   d. Proof that taxes, fees and similar obligations have been paid.
   e. Removal of temporary facilities and services.
   f. Removal of surplus materials, rubbish and similar elements.
   g. Release of claims.

D. Request for Information (RFI): If a question, information or clarification regarding any part of the Contract or FBOP Requirements is needed, the Contractor shall submit a Request for Information (RFI) to the Contracting Officer in accordance with the following RFI submission process:
   1. The Contractor shall submit a written, signed and dated RFI to the Contracting Officer.
   2. Each RFI shall be numbered sequentially using a unique series of numbers for document control purposes.
   3. The Contractor shall all a reasonable amount of time from the date of receipt, for the Contracting Officer's review and written reply to the RFI.

E. Contract Modifications (Change Orders): When changes are made in the work that requires an equitable adjustment to the Contract, the Contractor shall submit to the Contracting Officer a written proposal for adjustment of the amount of increase or decrease. Each Contract Modification Proposal (Change Order) shall be numbered sequentially using a unique series of numbers for document control purposes. With the proposal, the Contractor shall submit an itemized breakdown in at least the following detail:
   1. Material quantities and item cost.
   2. Labor costs by material item.
   3. Construction equipment cost.
   4. Workmen's compensation and public liability insurance.
   5. Overhead.
   6. Profit or commission.
   7. Employment taxes (FICA and FUTA); **no overhead, profit or commission will be allowed on employment taxes.**
   8 Request for time extensions (calendar days) if applicable.

   If the Contracting Officer determines that an equitable adjustment in time and dollars is to be made, the Contract will be modified in writing using the prescribed Stand Form provided by the Contracting Officer.

1.09 TEMPORARY CONSTRUCTION FACILITIES AND CONTROLS

A. The Contractor shall provide the following temporary provisions for the duration of the Contract.
   1. Temporary heating and ventilating as necessary to protect materials, products and finishes from damage due to temperature or humidity and to maintain specified conditions during construction.
   2. Temporary field office, with adequate electrical power, lighting, telephone, fax, heating and cooling to maintain human comfort.
   3. Temporary sanitary facilities. Existing and permanent toilets may not be used during construction.

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4. Temporary barriers, barricades and similar devices as necessary for safety and protection of construction personnel and public.
5. Temporary tree and plant protection to preserve existing trees and plants. Replace trees and plants damage by construction operations with new of equivalent caliper. Coordinate with COR.
6. Temporary utilities, as necessary to complete the contract:
   a. Water Service: Water from the FBOP institution's existing water system is available for use without metering and without payment of use charges. The Contractor shall provide connections and extensions of services as required for construction operations, and protect the temporary service from freezing. The Contractor shall exercise measures to conserve water, and regulate the temporary system to prevent interference with the Institution's normal usage.
   b. Electric Power Service: Electric power from the FBOP institution's existing system is available for use without metering and without payment of use charges. The Contractor shall provide connections and extensions of services as required for construction operations. The Contractor shall exercise measures to conserve electricity, and regulate the temporary system to prevent interference with the Institution's normal usage.
7. Temporary access roads and approaches to serve the construction area.
8. Temporary dust, erosion and sediment controls. These include keeping streets, drives and walks adjacent to the work site and haul routes clean and free of dirt, debris and litter caused by construction operations.

1.10 SAFETY AND HEALTH

A. The Contractor is responsible for safety and health on the project. In performing the work under this contract the Contractor shall:
   1. Conform to the specific safety requirements established by this contract.
   2. Comply with the safety rules of the FBOP institution that concern related activities not directly addressed in this contract.
   3. Take all reasonable steps and precautions to prevent accidents and preserve the life and health of Contractor and Government personnel performing or in any way coming in contact with the performance of this contract.
   4. Take such additional immediate precautions as the Contracting Officer may reasonably require for safety and accident prevention purposes.

1.11 CONSTRUCTION CLEANING

A. The Contractor shall execute continual cleaning to keep building, site and adjacent properties free from accumulations of waste materials, litter, debris, rubbish, windblown debris and foreign substances resulting from construction operations. The entire work area must be kept free from all debris. The soils must be free of nails, wires, re-bar and other construction materials. During and after the construction process there must be no burying of any construction debris.

B. Remove waste materials, debris, and rubbish from premises and dispose off-site in a lawful manner. Do not dispose of volatile waste such as mineral spirits, oil, paint thinner in storm or sanitary drains. Do not dispose of wastes into streams or waterways.

C. If a dispute arises between the Contractor and the Government as to the responsibility under this Contract for maintaining the premises and surrounding area free from waste materials, litter and so on; the Government may clean up; and the Contracting Officer will allocate the cost to the Contractor via modification to the Contract.

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1.12 SUBMITTALS FOR SUBSTITUTION

A. General Requirements: Unless otherwise specifically provided in this contract, reference to any equipment, material, article, or patented process, by trade name, make, or catalog number, shall be regarded as establishing a standard of quality and shall not be construed as limiting competition. Subject to compliance with specified requirements, the Contractor may, at its option, use any equipment, material, article, or process, which in the judgment of the Contracting Officer is equal to that named.

B. Substitutions for Cause: Changes proposed by the Contractor that are required due to changed Project conditions, such as unavailability of product, regulatory changes, or unavailability of required warranty items.

C. Substitutions for Convenience: Changes proposed by the Contractor that are not required in order to meet other Project requirements but may offer advantage to the Contractor or the Government.

D. Request for Substitutions shall be made on discovery of need for the change, but not later than 30 days prior to time required for preparation and review of related substitution submittals. Requests for substitutions shall be submitted to the Contracting Officer, and clearly marked for Cause or Convenience.
   1. Submit four (4) copies, fully identified for product or method being proposed for substitution, including related Specification Section and drawing number(s), and fully documented to show compliance with requirements for substitutions. Include project data/drawings, descriptions of methods, samples where applicable, detailed comparison of significant qualities between specified item and proposed substitution, statement of effect on construction time and coordination with other affected cost information or proposal, and statement to the effect that proposed substitution will result in overall work equal to or better than work originally indicated.
   2. Document each request with supporting data substantiating compliance of proposed substitution with contract documents, including:
      a. Manufacturer’s name and address, product, trade name, mode/catalog number, performance and test data and reference standards.
      b. Itemized point-by-point comparison of proposed substitution with specific product, listing variations in quality, performance and other pertinent characteristics.
      c. Changes required in other Work.
      d. Availability of maintenance service and source of replacement parts, as applicable.
      e. Certified test data to show compliance with performance characteristics specified.
      f. Samples, when applicable or requested.
      g. Other information as necessary to assist in Contracting Officer’s evaluation.
   3. Properly submitted substitutions that are acceptable to the Contracting Officer will be included in Contract Documents by modifications.
   4. Substitutions will not be considered for acceptance when:
      a. They are indicated or implied on shop drawings or product data submittals without a formal request from Contractor.
      b. Acceptance will require substantial revision of contract documents.
      c. Submittal for substitution request does not include point-by-point comparison of proposed substitution with specified product.
      d. In judgement of Contracting Officer, submittal does not include adequate information necessary for a complete evaluation.
      e. If requested directly by a subcontractor or supplier.

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5. Substitute products shall not be ordered or installed without written acceptance of Contracting Officer.
6. Contracting Officer will determine acceptability of proposed substitutions.
7. No substitutions are permitted after execution of the Contract, except as allowed by Contract Modification (Change Order).

1.13 QUALITY ASSURANCE

A. Source Limitations: To the greatest extent possible for each unit of Work, and subject to the restrictions of the Buy American Act, provide products, materials, or equipment of a singular generic kind from a single source.

B. Compatibility of Options: Where more than one choice is available an options for Contractor's selection of a product or material, select an option that is compatible with other products and materials already selected. Total compatibility among options is not assured by limitations within contract documents, but must be provided by Contractor. Compatibility is a basic general requirement of product/material selections.

1.14 PRODUCT DELIVERY/STORAGE/HANDLING

A. General: Deliver, handle, and store products in accordance with manufacturer's recommendations and by methods and means that will prevent damage, deterioration, and loss, including theft, and protect against damage from climatic conditions. Control delivery schedules to minimize long-term storage of products at site and overcrowding of construction spaces. In particular, provide delivery/installation coordination to ensure minimum holding or storage times for products recognized to be flammable, hazardous, easily damaged, or sensitive to deterioration, theft, and other sources of loss. Damaged or defective items, in the opinion of the COR, shall be replaced at no cost to the Government.

1.15 WARRANTIES (GUARANTEES)

A. General: The warranty and guarantee provisions of the Contract Clauses apply to all Work of the contract. In accordance with FAR 52.246.21 Warranty of Construction, and in addition to any other warranties in this contract, the Contractor warrants that work performed under this contract conforms to the contract requirements and is free of any defect in equipment, material, or workmanship performed by the contractor or any subcontractor or supplier at any tier. The warranty shall continue for a period of one (1) year from the date of Project Substantial Completion.

B. Refer to individual specification sections for the determination of units of Work that are required to be specifically or individually warranted, and for the specific requirements and terms of those warranties (or guarantees) in excess of what is required by the Contract Clauses.
1. Submit Special Project Warranty (Guarantee): A warranty specifically written and signed by Contractor for a defined portion of the Work and, where required, countersigned by subcontractor, installer, manufacturer, or other entity engaged by Contractor.
2. Specified Project Warranty: A warranty that is required by contract documents to be provided for a manufactured product incorporated into the Work, regardless of whether manufacturer has published a similar warranty without regard for specific incorporation of product into the Work, or has written and executed a special project warranty as a direct result of contract document requirements.
3. Coincidental Product Warranty: A warranty that is not specifically required by contract documents (other than as specified in these Requirements), but that is available on a product incorporated into the Work, by virtue of the fact that manufacturer of product has published warranty in connection with purchases and uses of product without regard for specific applications except as otherwise limited by terms of warranty.

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1.16 PROJECT CLOSEOUT

A. Substantial Completion: When the Contractor considers the work substantially complete, submit the following to the Contracting Officer.
   1. A written notice that work, or designated portion thereof, is substantially complete, and request an inspection by the COR and the Contracting Officer.
   2. A punch list of remaining incomplete/non-compliant items shall accompany this written notice. Prior to the Contractor requesting substantial completion, the Contractor shall develop a punch list of incomplete and non-compliant items.
   3. Within a reasonable time after receipt of such notice, the Contracting Officer and COR will make an inspection to determine the status of substantial completion.
   4. If the Contracting Officer determines that the work is not substantially complete:
      a. The Contracting Officer will promptly notify the Contractor in writing, giving reasons therefore.
      b. The Contractor shall remedy deficiencies in the work and send a second written notice of substantial completion to the Contracting Officer.
      c. The Contracting Officer and the COR will re-inspect the work.
      d. The Contractor shall be responsible for any additional inspections by the Government that are not fault of the Government.
   5. When the Contracting Officer determines that the work is substantially complete, the Contracting Officer will:
      a. Request the COR verify the Contractor's punch list. The Contractor's punch list may be amended by the COR with any open issues and any remaining items from the incomplete/non-compliant list.
      b. Request the Contractor to prepare a Certificate of Substantial Completion on a form approved by the Contracting Officer, accompanied by the punch list. This form will be used to record the established start dates for warranties and guaranties. This document provides for agreement as to the time allowed for completion or correction of work remaining, the date upon which the FBOP institution will assume possession of the work. If issued after the contract completion date, the certificate establishes the date upon which liquidated damages cease.

B. Final Inspection and Acceptance: The Contractor shall diligently complete all punch list items before a Final Inspection is scheduled.
   1. When the Contractor considers the work complete, he/she shall submit a written certification to the Contracting Officer stating the Work has been completed and inspected for compliance with the Contract Documents; and the Work is ready for final inspection. The Contractor shall give the Contracting Officer a minimum of ten (10) work days advanced written notice of the date on which the work will be fully completed and ready for the Government's final inspection.
   2. The Contracting Officer and the COR shall make an inspection to verify the status of completion with reasonable promptness after receipt of such certification.
   3. Should the Contracting Officer consider that the Work is incomplete or defective:
      a. The Contracting Officer will promptly notify the Contractor in writing, listing the incomplete or defective work.
      b. The Contractor shall take immediate steps to remedy the stated deficiencies and send a second written certification to the Contracting Officer when the work is complete.
      c. The Contracting Officer and the COR will re-inspect the work.
      d. The Contractor shall be responsible for any additional inspections by the Government that are not fault of the Government.

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e. The Contracting Officer shall assign monetary value to all punch list items not completed, and to all required submittals not received, as of the date of Final Acceptance. The total value of those items shall be retained and/or deducted from the Contractor's final payment until the Contractor demonstrates to the Contracting Officer's satisfaction that such items have been completed or corrected.

4. When the Contracting Officer finds that work is acceptable under the Contract Documents, the Contracting Officer will request the Contractor to complete all remaining closeout submittals and contract requirements, including but not limited to: GSA Form 1142 – Release of Claims, operating and maintenance manuals, record documents, warranties, guarantees, certificates, clearances, and bonds.
5. When all work is acceptable, the Contracting Officer will issue a Certificate of Contract Completion, GSA Form 218, or other approved form to the Contractor for signature.

C. Final Application for Payment: After the work is fully accepted by the Contracting Officer, the Contractor shall submit the final application for payment in accordance with procedures and requirements stated in the Contract requirements. As required by the Contracting Officer, the final application for payment shall include affidavits that all bills have been paid. The Contractor's submission of the final application for payment shall signify that the work is complete, and that the Contractor has vacated the Project work site

END OF SECTION

SECTION 02 4100
DEMOLITION

PART 1 GENERAL

1.01 SECTION INCLUDES

A. Building demolition excluding removal of hazardous materials and toxic substances.

B. Selective demolition of roofing systems.

C. Selective demolition of building elements for alteration purposes.

1.02 REFERENCE STANDARDS

A. 29 CFR 1926 - U.S. Occupational Safety and Health Standards; current edition.

B. NFPA 241 - Standard for Safeguarding Construction, Alteration, and Demolition Operations; 2013.

1.03 SUBMITTALS

A. Site Plan: Showing:
   1. Areas for temporary construction and field offices.
   2. Areas for temporary and permanent placement of removed materials.

B. Demolition Plan: Submit demolition plans.
   1. Indicate extent of demolition, removal sequence, bracing and shoring, and location and construction of barricades and fences.
   2. Permit for transportation & disposal of debris.

C. Project Record Documents: Accurately record actual locations of capped and active utilities and subsurface construction.

PART 3 EXECUTION

2.01 SCOPE

A. Remove extent of roofing as indicated on drawings.

B. Remove other items indicated, for salvage, relocation, and recycling.

2.02 GENERAL PROCEDURES AND PROJECT CONDITIONS

A. Comply with applicable codes and regulations for demolition operations and safety of adjacent structures and the public.
   1. Obtain required permits.
   2. Comply with applicable requirements of NFPA 241.
   3. Use of explosives is not permitted.
   4. Take precautions to prevent catastrophic or uncontrolled collapse of structures to be removed; do not allow worker or public access within range of potential collapse of unstable structures.
   5. Provide, erect, and maintain temporary barriers and security devices.
   6. Conduct operations to minimize effects on and interference with adjacent structures and occupants.
   7. Do not close or obstruct roadways or sidewalks without permit.
   8. Conduct operations to minimize obstruction of public and private entrances and exits; do not obstruct required exits at any time; protect persons using entrances and exits from removal operations.
   9. Obtain written permission from owners of adjacent properties when demolition equipment will traverse, infringe upon or limit access to their property.

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B. Do not begin removal until receipt of notification to proceed from COR.

C. Protect existing structures and other elements that are not to be removed.
   1. Provide bracing and shoring.
   2. Prevent movement or settlement of adjacent structures.
   3. Stop work immediately if adjacent structures appear to be in danger.

D. If hazardous materials are discovered during removal operations, stop work and notify COR; hazardous materials include regulated asbestos containing materials, lead, PCB's, and mercury.

2.03 EXISTING UTILITIES

A. Coordinate work with utility companies; notify before starting work and comply with their requirements; obtain required permits.

B. Protect existing utilities to remain from damage.

C. Do not close, shut off, or disrupt existing life safety systems that are in use without at least 7 days prior written notification to COR.

D. Do not close, shut off, or disrupt existing utility branches or take-offs that are in use without at least 3 days prior written notification to COR.

E. Locate and mark utilities to remain; mark using highly visible tags or flags, with identification of utility type; protect from damage due to subsequent construction, using substantial barricades if necessary.

2.04 SELECTIVE DEMOLITION FOR ALTERATIONS

A. Separate areas in which demolition is being conducted from other areas that are still occupied.
   1. Provide, erect, and maintain temporary dustproof partitions of construction in locations indicated on drawings.

B. Maintain weatherproof exterior building enclosure except for interruptions required for replacement or modifications; take care to prevent water and humidity damage.

C. Remove existing work as indicated and as required to accomplish new work.
   1. Remove items indicated on drawings.

D. Services (Including but not limited to HVAC, Plumbing, Fire Protection, Electrical, and Telecommunications): Remove existing systems and equipment as indicated.
   1. Maintain existing active systems that are to remain in operation; maintain access to equipment and operational components.
   2. Where existing active systems serve occupied facilities but are to be replaced with new services, maintain existing systems in service until new systems are complete and ready for service.
   3. Verify that abandoned services serve only abandoned facilities before removal.
   4. Remove abandoned pipe, ducts, conduits, and equipment, including those above accessible ceilings; remove back to source of supply where possible, otherwise cap stub and tag with identification.

E. Protect existing work to remain.
   1. Prevent movement of structure; provide shoring and bracing if necessary.
   2. Perform cutting to accomplish removals neatly and as specified for cutting new work.
   3. Repair adjacent construction and finishes damaged during removal work.
   4. Patch as specified for patching new work.

2.05 DEBRIS AND WASTE REMOVAL

A. Remove debris, junk, and trash from site.

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B. Leave site in clean condition, ready for subsequent work.
C. Clean up spillage and wind-blown debris from public and private lands.
D. Transport to offsite location and legally dispose of debris.

END OF SECTION

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SECTION 07 0150.19

PREPARATION FOR RE-ROOFING

PART 1 GENERAL

1.01 SECTION INCLUDES

A. Replacement of existing roofing system in preparation for new roofing system or partial replacement.

B. Partial replacement of existing roofing system in preparation for replacement roofing system in designated areas as indicated on drawings.

C. Removal of existing flashing and counterflashings.

D. Temporary roofing protection.

1.02 REFERENCE STANDARDS

A. ASTM D2178/D2178M - Standard Specification for Asphalt Glass Felt Used in Roofing and Waterproofing; 2015a.

B. ASTM D312/D312M - Standard Specification for Asphalt Used in Roofing; 2016a.

C. ASTM D41/D41M - Standard Specification for Asphalt Primer Used in Roofing, Dampproofing, and Waterproofing; 2011 (Reapproved 2016).

D. ASTM D4601/D4601M - Standard Specification for Asphalt-Coated Glass Fiber Base Sheet Used in Roofing; 2004, with Editorial Revision (2012).

E. PS 1 - Structural Plywood; 2009.

1.03 ADMINISTRATIVE REQUIREMENTS

A. Coordinate with affected mechanical and electrical work associated with roof penetrations.

B. Preinstallation Meeting: Convene one week before starting work of this section.
   1. Attendees:
      a. COR.
      b. Contractor.
      c. Installer.
      d. Roofing system manufacturer's field representative.
      e. Inspection and Testing Agency Representatives.
   2. Meeting Agenda: Provide agenda to participants prior to meeting in preparation for discussions on the following:
      a. Removal and installation schedule.
      b. Necessary preparatory work.
      c. Protection before, during, and after roofing system installation.
      d. Assure the accuracy and completeness of the roofing submittal. Correct any deficiencies and omissions, then resubmit the required information as required.
      e. Structural loading limitations of roof deck during re-roofing.
      f. Base flashing, special roofing details, drainage, penetrations, equipment curbs, and condition of other construction that affect re-roofing.
      g. Removal of existing roofing system.
      h. Existing roof deck conditions.
      i. Existing roof deck removal procedures and COR notifications.
      j. Condition and acceptance of existing roof deck and base flashing substrate for reuse.
      k. Asbestos removal and discovery of asbestos - containing material.
      l. Installation of new roofing system.
      m. Temporary roofing and daily terminations.

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n. Transitions and connection to and with other work.
o. Inspections and testing of installed systems.
p. Review the Total Roofing System Warranty and certifications.
q. Document the meeting and distribute copies of the minutes to all the attending parties.
r. Review of site logistics plan and secure site access.

C. Schedule work to coincide with commencement of installation of new roofing system.

1.04 SUBMITTALS

A. Site logistics plan. Include the following into the plan:
1. Construction staging area
2. Dumpster locations
3. Parking
4. Worker access
5. Truck access
6. Access routes
7. Check-in and check-out procedures
8. Temporary construction fencing
9. Protection of adjoining Property
10. Tree protection
11. Construction sign
12. Material storage
13. Fire extinguishers
14. Field restroom, portable or use of existing facilities.
15. Hydrant meter
16. Crane staging
17. Utility access
18. Construction hours
19. Temporary field office

B. Product Data: Submit for each type of material.

C. Shop Drawings: Indicate size, configuration, and installation details.

D. Photographs or Videotape of existing conditions.

E. Installer's Qualification Statement.

1.05 CLOSEOUT SUBMITTALS

A. Maintenance Data: For roofing system to include in maintenance manuals.

B. Certified statement from existing roof membrane manufacturer, stating that existing roof warranty has not been affected by Work performed under this Section.

1.06 QUALITY ASSURANCE

A. Installer Qualifications: Company specializing in performing work of the type specified and with at least five years of documented experience.
1. Approved by existing roofing system warrantor to work on existing warranted roof system.

1.07 DELIVERY, STORAGE, AND HANDLING

A. Ensure storage and staging of materials does not exceed static and dynamic load-bearing capacities of roof decking.

B. Store materials susceptible to weather in dry locations.

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1.08 FIELD CONDITIONS

A. Existing Roofing System: Modified bitumen, single ply or other existing roofing system.

B. Do not remove existing roofing membrane when weather conditions threaten the integrity of building contents or intended continued occupancy.

C. Maintain continuous temporary protection prior to and during installation of new roofing system.

D. COR will occupy building areas directly below re-roofing area.
   1. Provide COR with at least 2 weeks written notice of roofing activities that may affect their operations and to allow them to prepare for upcoming activities as necessary.
   2. Do not disrupt COR's operations or activities.
   3. Maintain access of COR's personnel to corridors, existing walkways, and adjacent buildings.

E. Review existing conditions to insure the drainage, layout of the drains and quantity are adequate for the re-roofing project.

F. Review existing gutter, scuppers and downspouts are re-usable and can support drainage with the re-roofing system.

G. Review scuppers flashing and sealants for compatibility, damage, repairs, etc.

H. Review existing splashblocks for re-use or replacement.

I. Review limit of construction loads on existing roof.

J. Weather Limitations: Proceed with re-roofing preparation only when existing and forecasted weather conditions permit Work to proceed without water entering existing roofing system or building.
   1. Remove only as much roofing in one day as can be watertight in the same day.

K. Hazardous Materials: It is not expected that hazardous materials, such as asbestos-containing materials, will be encountered in the Work.
   1. Hazardous materials will be removed by COR before start of the Work.
   2. Existing roof will be left no less watertight than before removal.
   3. If materials suspected of containing hazardous materials are encountered, do not disturb; immediately notify COR.
      a. Hazardous materials will be removed by COR under a separate contract.

1.09 WARRANTY

A. Existing Warranties: Perform this work using methods and materials that will maintain existing roof system warranties.
   1. Notify existing roof system warrantor prior to starting this work and obtain written instructions for procedures necessary to maintain this existing warranty.
   2. Upon completion of this work, notify warrantor of re-roofing completion and obtain documentation to verify that existing roofing system has been inspected and warranty is still in effect.
      a. Submit documentation upon project closeout.

PART 2 PRODUCTS

2.01 MATERIALS

A. Patching Materials: Provide necessary materials in accordance with requirements of existing roofing system.

B. Temporary Roofing Protection Materials:
   1. Contractor's responsibility to select appropriate materials for temporary protection of roofing areas as determined necessary for this work.

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2.02 ACCESSORIES

A. Fasteners: Type and size as required and compatible with existing and new roofing system to resist local wind uplift.

B. Sheathing Paper: Red rosin paper type, at least 3 lbs per 100 sq ft.

C. Base Sheet: Non-perforated, asphalt-coated glass fiber base sheet, Type II in accordance with ASTM D4601/D4601M.

D. Glass Fiber Felt: Asphalt-impregnated, glass fiber felt, Type IV in accordance with ASTM D2178/D2178M.

E. Asphalt Primer: Apply to concrete decking in preparation for temporary roofing, Type II in accordance with ASTM D41/D41M.

F. Roofing Asphalt: Type III or Type IV, in accordance with ASTM D312/D312M.

PART 3 EXECUTION

3.01 EXAMINATION

A. Examine substrates, area, and conditions, with manufacturer's representative and installer present, for compliance with the following requirements and other conditions affecting performance of roofing system:
   1. Verify that roof openings and penetrations are in place and curbs are set and braced and that roof drain bodies are securely clamped in place.
   2. Verify that wood blocking, curbs, and nailers are securely anchored to roof deck at penetrations and terminations and that nailers match thickness of insulation.
   3. Verify existing roof surface is clear and ready for work in this section.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.02 PREPARATION

A. Protection of In-Place Conditions:
   1. Protect existing roofing system that is not to be re-roofed.
   2. Limit traffic and material storage to areas of existing roofing that have been protected.
   3. Maintain temporary protection and leave in place until replacement roofing has been completed. Remove temporary protection on completion of re-roofing.
   4. Comply with requirements of existing roof system manufacturer's warranty requirements.

B. Coordinate with COR to shut off rooftop utilities and service piping before beginning the Work.

C. Test existing roof drains to verify that they are not blocked or restricted.
   1. Immediately notify COR of any blockage or restrictions.

D. Coordinate with COR to shut down air-intake equipment in the vicinity of the Work.
   1. Cover air-intake louvers before proceeding with re-roofing work that could affect indoor air quality or activate smoke detectors in the ductwork.

E. During removal operations, have sufficient and suitable materials on-site to facilitate rapid installation of temporary protection in the event of unexpected rain.

F. Maintain roof drains in functioning condition to ensure roof drainage at end of each workday.
   1. Prevent debris from entering or blocking roof drains and conductors.
      a. Use roof drain plugs specifically designed for this purpose.
      b. Remove roof-drain plugs at end of each workday, when no work is taking place, or when rain is forecast.
   2. If roof drains are temporarily blocked or unserviceable due to roofing system removal or partial installation of new roofing system, provide alternative drainage method to remove water and eliminate ponding.

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a. Do not permit water to enter into or under existing roofing system components that are to remain.

G. Clean substrate of dust, debris, moisture, and other substances detrimental to roofing installation according to roofing system manufacturer's written instructions. Remove sharp projections.

H. Complete terminations and base flashings and provide temporary seals to prevent water from entering completed sections of roofing system at the end of the workday or when rain is forecast. Remove and discard temporary seals before beginning work on adjoining roofing.

I. Remove loose refuse and dispose of properly off-site.

3.03 ROOF TEAR-OFF

A. Notify COR each day of extent of roof tear-off proposed for that day.

B. Lower removed roofing materials to ground and onto lower roof levels, using dust-tight chutes or other acceptable means of removing materials from roof areas.

C. Remove aggregate ballast from roofing.

D. Remove loose aggregate from aggregate-surfaced, built-up bituminous roofing using a power broom.

E. Remove pavers and accessories from roofing.
1. Store and protect pavers and accessories for reuse in manner not to exceed structural loading limitations of roof deck.
2. Discard cracked pavers.

F. Remove ballast, protection mat, and insulation from protected roofing membrane.
1. Discard insulation that is damaged.
2. Store insulation for reuse and protect it from physical damage.
3. Store ballast for reuse in manner not to exceed structural loading limitations of roof deck.

G. Full Roof Tear-off: Where indicated on Drawings, remove existing roofing and other roofing system components down to the existing roof deck.
1. Remove substrate board, vapor retarder, roof insulation and cover board.
2. Remove base flashings and counter flashings.
3. Remove perimeter edge flashing and gravel stops.
4. Remove copings.
5. Remove expansion-joint covers.
6. Remove flashings at pipes, curbs, mechanical equipment, and other penetrations.
7. Remove roof drains indicated on Drawings to be removed.
8. Remove wood blocking, curbs, and nailers.
9. Bitumen and felts that are firmly bonded to concrete decks are permitted to remain if felts are dry.
   a. Remove unadhered bitumen, unadhered felts, and wet felts.
10. Remove excess asphalt from steel deck.
11. Remove fasteners from deck.

H. Partial Roof Tear-off: Where indicated on Drawings, remove existing roofing down depth noted and immediately check for presence of moisture.
1. Testing Agency: COR will engage a qualified testing agency to inspect substrate conditions, surface preparation, roof membrane application, flashings, protection, and drainage components, and to furnish reports back to the COR.
2. Perform the following test:

a. Survey exposed substrate that is to remain using infrared color thermography according to ASTM C1153.
1) Prepare survey report of initial scan indicating locations of entrapped moisture, if any, and area calculations of locations of entrapped moisture.

3. Remove wet or damp materials below existing roofing and above deck as directed by COR.
4. Inspect wood blocking, curbs, and nailers for deterioration and damage.
a. If wood blocking, curbs, or nailers have deteriorated, immediately notify COR.
5. Bitumen and felts that are firmly bonded to concrete decks are permitted to remain if felts are dry.
a. Remove unadhered bitumen, unadhered felts, and wet felts.
6. Remove excess asphalt from steel deck that is exposed by removal of wet or damp materials.
a. A maximum of 15 lb/100 sq. ft. (0.72 kg/sq. m) of asphalt is permitted to remain on steel decks.
7. Remove fasteners from deck

3.04 DECK PREPARATION

A. Inspect deck after tear-off of roofing system.

B. If deck surface is unsuitable for receiving new roofing or if structural integrity of deck is suspect, immediately notify COR.
1. Do not proceed with installation until directed by COR.

3.05 INFILL MATERIALS INSTALLATION

A. Immediately after roof tear-off, and inspection and repair, if needed, of deck, fill in tear-off areas to match existing roofing system construction.

B. Install new roofing patch over roof infill area.
1. If new roofing is installed the same day tear-off is made, roofing patch is not required.

3.06 BASE FLASHING REMOVAL

A. Remove existing base flashings.
1. Clean substrates of contaminants, such as asphalt, sheet materials, dirt, and debris.

B. Do not damage metal counterflashings that are to remain.
1. Replace metal counterflashings damaged during removal with counterflashings of same metal, weight or thickness, and finish as existing.

C. Inspect parapet sheathing, wood blocking, curbs, and nailers for deterioration and damage.
1. If parapet sheathing, wood blocking, curbs, or nailers have deteriorated, immediately notify COR.

3.07 FASTENER PULL-OUT TESTING

A. Perform fastener pull-out tests according to SPRI FX-1, and submit test report to COR before installing new roofing system.

3.08 DISPOSAL

A. Collect demolished materials and place in containers.
1. Promptly dispose of demolished materials.
2. Do not allow demolished materials to accumulate on-site.
3. Storage or sale of demolished items or materials on-site is not permitted.

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3.09 FIELD QUALITY CONTROL

A. Testing Agency: COR will engage a qualified testing agency to inspect substrate conditions, surface preparation, roof membrane application, flashings, protection, and drainage components, and to furnish reports back to the COR.
   1. Coordinate with COR's testing agency to schedule times for tests and inspections immediately after removal.

B. Flood testing: In geographic areas where roofs require water retention capability, flood test each deck area for leaks according to recommendations in ASTM C 5957 after completing waterproofing but before overlaying construction is in place.
   1. Flood test to an average depth of 65mm (2 1/2") with a minimum depth of 25mm (1") and not exceeding 100mm (4")
   2. Floor test each area for 48 hours.
   3. After testing, repair leaks and repeat flood testing. Make repairs and repeat testing until area is watertight.
   4. Test critical sheet flashing by localized water testing. Make repairs and retest until flashing is watertight.

C. Testing will identify the condition of existing materials and their reuse, repair or removal.

D. Test Reports: Indicate existing insulation moisture content and existing bitumen quality.

3.10 FINAL INSPECTIONS

A. Final Roof Inspection: Arrange for roofing system manufacturer's technical personnel to inspect roofing installation on completion and provide a punchlist as required.

B. Repair or remove and replace components of membrane roofing system where inspections indicate that they do not comply with specified requirements.

C. Follow-up and perform any corrective work described in the punchlist from the manufacturer's technical representative.

D. Additional inspections, at Contractor's expense will be performed to determine compliance of replaced or additional work with specified requirements.

END OF SECTION

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SECTION 07 4113

METAL ROOF PANELS

PART 1 GENERAL

1.01 SECTION INCLUDES

A. Architectural roofing system of preformed steel panels.

B. Architectural roofing system of preformed aluminum panels.

C. Thermal roof insulation.

D. Attachment system.

E. Roof Repair

F. Roof Replacement

G. Accessories.

1.02 REFERENCE STANDARDS

A. ASCE 7 - Minimum Design Loads and Associated Criteria for Buildings and Other Structures; Most Recent Edition Cited by Referring Code or Reference Standard.

B. ASTM A463/A463M - Standard Specification for Steel Sheet, Aluminum-Coated, by the Hot-Dip Process; 2015.

C. ASTM A653/A653M - Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process; 2018.

D. ASTM A792/A792M - Standard Specification for Steel Sheet, 55% Aluminum-Zinc Alloy-Coated by the Hot-Dip Process; 2010 (Reapproved 2015).

E. ASTM A1011/A1011M - Standard Specification for Steel, Sheet and Strip, Hot-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, and Ultra-High Strength; 2018a.

F. ASTM B209 - Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate; 2014.

G. ASTM C1363 - Standard Test Method for Thermal Performance of Building Materials and Envelope Assemblies by Means of a Hot Box Apparatus; 2011.

H. ASTM D1970/D1970M - Standard Specification for Self-Adhering Polymer Modified Bituminous Sheet Materials Used as Steep Roofing Underlayment for Ice Dam Protection; 2018.

I. ASTM E84 - Standard Test Method for Surface Burning Characteristics of Building Materials; 2018b.

J. ASTM E96/E96M - Standard Test Methods for Water Vapor Transmission of Materials; 2016.

K. ASTM E1592 - Standard Test Method for Structural Performance of Sheet Metal Roof and Siding Systems by Uniform Static Air Pressure Difference; 2005 (Reapproved 2017).

L. ASTM E1646 - Standard Test Method for Water Penetration of Exterior Metal Roof Panel Systems by Uniform Static Air Pressure Difference; 1995 (Reapproved 2018).

M. ASTM E1680 - Standard Test Method for Rate of Air Leakage Through Exterior Metal Roof Panel Systems; 2016.

N. UL 580 - Standard for Tests for Uplift Resistance of Roof Assemblies; Current Edition, Including All Revisions.

1.03 ADMINISTRATIVE REQUIREMENTS

A. Preinstallation Meeting: Convene one week before starting work on this section.

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1. Review preparation and installation procedures and coordinating and scheduling required with related work.
2. Meet with COR, COR's insurer if applicable, metal panel installer, metal panel manufacturer's representative, structural-support installer, and installers whose work interfaces with or affects metal panels, including installers of roof accessories and roof-mounted equipment.
3. Review and finalize construction schedule and verify availability of materials, installer's personnel, equipment, and facilities needed to make progress and avoid delays.
4. Review methods and procedures related to metal panel installation, including manufacturer's written instructions.
5. Examine support conditions for compliance with requirements, including alignment between and attachment to structural members.
6. Review structural loading limitations of deck, purlins and rafters during and after roofing.
7. Review flashings, special details, drainage, penetrations, equipment curbs, and condition of other construction that affect metal panels.
8. Review governing regulations and requirements for insurance, certificates, and tests and inspections if applicable.
9. Review temporary protection requirements for metal panel systems during and after installation.
10. Review procedures for repair of metal panels damaged after installation.
11. Document proceedings, including corrective measures and actions required, and furnish copy of record to each participant.

1.04 SUBMITTALS

A. Product Data: Manufacturer's data sheets on each product to be used, including:
   1. Summary of test results, indicating compliance with specified requirements.
   2. Storage and handling requirements and recommendations.
   3. Installation methods.
   4. Specimen warranty.

B. Shop Drawings: Include layouts of roof panels, details of edge and penetration conditions, spacing and type of connections, flashings, underlayments, and special conditions.
   1. Show work to be field-fabricated or field-assembled.

C. Warranty: Submit specified manufacturer's warranty and ensure that forms have been completed in Owner's name and are registered with manufacturer.

1.05 CLOSEOUT SUBMITTALS

A. Maintenance Data: For metal panels to include in maintenance manuals.

1.06 QUALITY ASSURANCE

A. Manufacturer Qualifications: Company specializing in manufacturing products specified in this section, with not less than five years of documented experience.

B. Installer Qualifications: Company specializing in performing work of the type specified and with at least five years of documented experience.

1.07 DELIVERY, STORAGE, AND HANDLING

A. Provide strippable plastic protection on prefinished roofing panels for removal after installation.

B. Store roofing panels on project site as recommended by manufacturer to minimize damage to panels prior to installation.

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1.08 FIELD CONDITIONS

A. Weather Limitations: Proceed with installation only when existing and forecasted weather conditions permit assembly of metal panels to be performed according to manufacturers' written instructions and warranty requirements.

1.09 WARRANTY

A. Special Warranty: Manufacturer's standard form in which manufacturer agrees to repair or replace components of metal panel systems that fail in materials or workmanship within specified warranty period.
   1. Failures include, but are not limited to, the following:
      a. Structural failures including rupturing, cracking, or puncturing.
      b. Deterioration of metals and other materials beyond normal weathering.
   2. Warranty Period: 2 years from date of Substantial Completion.

B. Special Warranty on Panel Finishes: Manufacturer's standard form in which manufacturer agrees to repair finish or replace metal panels that show evidence of deterioration of factory-applied finishes within specified warranty period.
   1. Exposed Panel Finish: Deterioration includes, but is not limited to, the following:
      a. Color fading more than 5 Delta E units when tested according to ASTM D2244.
      b. Chalking in excess of a No. 8 rating when tested according to ASTM D4214.
      c. Cracking, checking, peeling, or failure of paint to adhere to bare metal.
   2. Finish Warranty Period: 20 years from date of Substantial Completion.

C. Special Weathertightness Warranty: Manufacturer's standard form in which manufacturer agrees to repair or replace standing-seam metal roof panel assemblies that fail to remain weathertight, including leaks, within specified warranty period.
   1. Warranty Period: 20 years from date of Substantial Completion.

D. Provide a roof system warranty as shown in Attachment: Roofing Installer’s Warranty 07 4113 Metal Roof Panels

PART 2 PRODUCTS

2.01 MANUFACTURERS

A. Metal Roof Panels:
   1. Berridge Manufacturing Company: www.berridge.com/#sle.
   2. Firestone Building Products LLC: www.firestonebpco.com/#sle.
   3. Metl-Span, a Division of NCI Group, Inc: www.metlspan.com/#sle.
   4. Centria: www.centria.com
   5. Approved Equal

2.02 ARCHITECTURAL METAL ROOF PANELS

A. Architectural Metal Roofing: Provide complete engineered system complying with specified requirements and capable of remaining weathertight while withstanding anticipated movement of substrate and thermally induced movement of roofing system.

B. Metal Panels: Factory-formed panels with factory-applied finish.
   1. Steel Panels:
      a. Aluminum-zinc alloy-coated steel complying with ASTM A792/A792M; minimum AZ50 coating.
      b. Steel Thickness: Minimum 24 gage (0.024 inch).
   2. Aluminum Panels:
      a. Alloy and Temper: Aluminum complying with ASTM B209 (ASTM B209M); temper as required for forming.

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b. Thickness: Minimum 20 gage (0.032 inch).
3. Texture: Smooth.
4. Length: Maximum possible length to minimize lapped joints. Where lapped joints are unavoidable, space laps so that each sheet spans over three or more supports.
5. Width: Maximum panel coverage of 24 inches.

2.03 ATTACHMENT SYSTEM

A. Concealed System: Provide manufacturer's standard stainless steel or nylon-coated aluminum concealed anchor clips designed for specific roofing system and engineered to meet performance requirements, including anticipated thermal movement.

2.04 FABRICATION

A. Panels: Provide factory or field fabricated panels with applied finish and accessory items, using manufacturer's standard processes as required to achieve specified appearance and performance requirements.

B. Joints: Provide captive gaskets, sealants, or separator strips at panel joints to ensure weathertight seals, eliminate metal-to-metal contact, and minimize noise from panel movements.

2.05 FINISHES

A. Fluoropolymer Coil Coating System: Manufacturer's standard multi-coat aluminum coil coating system complying with AAMA 2605, including at least 70 percent polyvinylidene fluoride (PVDF) resin, and at least 80 percent of coil coated aluminum surfaces having minimum total dry film thickness (DFT) of 0.9 mil, 0.0009 inch; color and gloss to match sample.

2.06 ACCESSORIES

A. Miscellaneous Sheet Metal Items: Provide flashings, gutters, downspouts, trim, moldings, closure strips, preformed crickets, caps, and equipment curbs of the same material, thickness, and finish as used for the roofing panels. Items completely concealed after installation may optionally be made of stainless steel.
1. Downspouts: match existing style

B. Rib and Ridge Closures: Provide prefabricated, close-fitting components of closed-cell synthetic rubber, neoprene, or PVC.

C. Sealants:
1. Exposed Sealant: Elastomeric; silicone, or polyurethane.
2. Concealed Sealant: Non-curing butyl sealant or tape sealant.
3. Seam Sealant: Factory-applied, non-skinning, non-drying type.

D. Snowguards: Provide snow guards for metal roofs having a slope equal to or greater than 2:12, fo projects anticipating ground snow loads equal to or greater than 0.5 kPa (10 psf).
1. Type: Rail-type, seam-mounted snow guards on sloped metal roof panel.
2. Include calculation of number and location of snow guards based on snow load, roof slope, roof type, components, spacings, and finish.
3. Preformance Requirements: Provide snow guards that withstand exposure to weather and resist thermally induced movement without failure, rattling, or fastener disengagement due to defective manufacture, fabrication, installation, or other defects in construction.
a. Temperature changes of 120 deg F, ambient; 180 deg F, material surfaces.
4. Material and Finish: Mill Finish Aluminum.

E. Thermal Insulation: Provide flexible blanket, rigid, or semi-rigid type, faced with white, flexible, non-dusting vapor retarder tested for maximum flame spread index of 50, per ASTM E84; for installation using spacer blocks.

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1. Thickness: As required to meet required thermal resistance.

F. Underlayment: Self-adhering rubber-modified asphalt sheet complying with ASTM D1970/D1970M; 22 mil total thickness; with strippable release film and woven polypropylene sheet top surface.
   1. Sheet Thickness: 22 mil, 0.022 inch minimum total thickness.
   2. Self Sealability: Passing nail sealability test specified in ASTM D1970/D1970M.
   3. Low Temperature Flexibility: Passing test specified in ASTM D1970/D1970M.
   4. Water Vapor Permeance: 0.1 perm, maximum, when tested in accordance with ASTM E96/E96M Procedure A (desiccant method).

PART 3 EXECUTION

3.01 EXAMINATION

A. Do not begin installation of preformed metal roof panels until substrates have been properly prepared.

B. If substrate preparation is the responsibility of another installer, notify COR of unsatisfactory preparation before proceeding.

3.02 PREPARATION

A. Coordinate roofing work with provisions for roof drainage, flashing, trim, penetrations, and other adjoining work to assure that the completed roof will be free of leaks.

B. Remove protective film from surface of roof panels immediately prior to installation. Strip film carefully, to avoid damage to prefinished surfaces.

C. Separate dissimilar metals by applying a bituminous coating, self-adhering rubberized asphalt sheet, or other permanent method approved by roof panel manufacturer.

D. Where metal will be in contact with wood or other absorbent material subject to wetting, seal joints with sealing compound and apply one coat of heavy-bodied bituminous paint.

3.03 INSTALLATION

A. Overall: Install roofing system in accordance with approved shop drawings and panel manufacturer's instructions and recommendations, as applicable to specific project conditions. Anchor all components of roofing system securely in place while allowing for thermal and structural movement.
   1. Install roofing system with concealed clips and fasteners, except as otherwise recommended by manufacturer for specific circumstances.
   2. Minimize field cutting of panels. Where field cutting is absolutely required, use methods that will not distort panel profiles. Use of torches for field cutting is absolutely prohibited.

B. Accessories: Install all components required for a complete roofing assembly, including flashings, gutters, downspouts, trim, moldings, closure strips, preformed crickets, caps, equipment curbs, rib closures, ridge closures, and similar roof accessory items.

C. Roof Panels: Install panels in strict accordance with manufacturer's instructions, minimizing transverse joints except at junction with penetrations.
   1. Form weathertight standing seams incorporating concealed clips, using an automatic mechanical seaming device approved by the panel manufacturer.

D. Insulation: Install insulation between roof covering and supporting members to present a neat appearance. Fold, staple, and tape seams unless otherwise approved by COR.

3.04 FIELD QUALITY CONTROL

A. Manufacturer's Field Service: Engage a factory-authorized service representative to test and inspect metal roof panel installation, including accessories. Report results in writing.

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B. Remove and replace applications of metal roof panels where tests and inspections indicate that they do not comply with specified requirements.

C. Prepare test and inspection reports.

3.05 CLEANING

A. Clean exposed sheet metal work at completion of installation. Remove grease and oil films, excess joint sealer, handling marks, and debris from installation, leaving the work clean and unmarked, free from dents, creases, waves, scratch marks, or other damage to the finish.

3.06 PROTECTION

A. Do not permit storage of materials or roof traffic on installed roof panels. Provide temporary walkways or planks as necessary to avoid damage to completed work. Protect roofing until completion of project.

B. Touch-up, repair, or replace damaged roof panels or accessories before Date of Substantial Completion.

END OF SECTION

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